		A V
22	**B**	Resistance of each of the lamp : $R = \dfrac{V^2_{rating}}{P_{rating}}$ since, both of them have the same voltage rating: $\dfrac{R_A}{R_B} = \dfrac{P_{rating,B}}{P_{rating,A}} = \dfrac{40}{10} = \dfrac{4}{1}$ Since, both lamps are connected in series, the same current passes both lamps and power emitted by each lamp : $P_{emitted} = I^2 R$. $\dfrac{P_{emitted,A}}{P_{emitted,B}} = \dfrac{R_A}{R_B} = \dfrac{4}{1} \Rightarrow P_A = 4P_B$

23	**A**	Consider the secondary circuit: when temperature is low the resistance of the thermistor is high. This means